

HARSHIL HK

Software Engineer & Startup Founder | Building AI-Powered Developer Tools | Full-Stack · Cloud · DevOps

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PROFESSIONAL SUMMARY

Software Engineer and startup founder with hands-on experience in **full-stack development**, **cloud infrastructure**, and **AI-powered applications**. Co-founded Evolune EdgeTech, developing Evo-TFX — an **AI-driven API testing platform** with 15+ beta teams and 9,000+ tests executed. Proven ability to ship real products in both startup and enterprise environments.

WORK EXPERIENCE

Associate Cloud Development Engineer | Full-time

Aug 2025 - Present

Bristlecone India

Remote • Bengaluru India

- **Developed** and **maintained AWS Lambda** functions for production-grade shipment tracking systems, diagnosing failures and deploying fixes via Jenkins CI/CD pipelines.
- **Analyzed** and resolved production tickets, including root cause investigation of data mismatch issues (e.g., Estes tracking ID format conflicts in the NEO system) with cross-team stakeholder communication.
- **Upgraded** QA environments to higher versions and replicated production Lambda configurations to QA for consistent and reliable testing parity.
- **Achieved** execution speed **20%** faster than peers.

Technologies / Skills Used : AWS, Python, Java, Node, PostgreSQL, React

Founding Engineer & SDE Intern | Self-employed

Jan 2024 - Mar 2025

Evolune EdgeTech

Remote • Bengaluru India

- Built and shipped **4 AI products** from scratch StyleSenseAI, Cal Coach, DarkPearl, and Flasqo spanning multimodal AI, NLP, and automated testing.
- Architected multimodal inference pipelines on Gemini, designed conversational API layers, and structured prompt engineering for **production-grade AI apps serving early paid subscribers**.
- Led test execution engine for Flasqo supporting **parallel runs across 3+ environments**, schema-driven API contract validation selected for **IIT Madras I-Summit PitchArena 2026**.
- Designed stateless, modular backends with serverless-optimized architecture reduced integration overhead by **40%+**, built for third-party and enterprise API scalability from day one.

Technologies / Skills Used : React, Python, Docker, Linux, PostgreSQL, Svelte, FastAPI, Microservices

PROJECTS

Flasqo | 🌐 Website

Jan 2024 - Jul 2025

Founding Engineer

- Built and launched a production SaaS platform that automates API test suite generation using OpenAI GPT-4, supporting 10+ testing types (functional, smoke, performance, chaos, contract, GraphQL, regression) reducing manual test writing by 90%.
- Engineered a modular microservice architecture with 5 self-contained FastAPI service modules, 120+ async REST endpoints, streaming responses for real-time test execution, and concurrent parallel test runs using httpx AsyncClient and Playwright browser automation.
- Deployed on Contabo VPS with Nginx reverse proxy, Gunicorn + Uvicorn (4 workers), PostgreSQL (connection pooling), Redis caching, SSL/TLS via Let's Encrypt, systemd service management, and UFW firewall for a hardened production environment.
- Implemented two-tier rate limiting (60 req/min general, 5 req/min auth), HMAC-SHA256 signed webhooks, SHA-256 hashed API keys, JWT authentication (HS256), OAuth2 (Google + GitHub), bcrypt password hashing, and security headers across the full stack.
- Integrated GitHub Actions CI/CD pipeline for zero-downtime automated deployments, GitHub webhook-based test triggers with commit status reporting, and a CI/CD API key system for pipeline-integrated test execution.

Technologies / Tools Used : Python, React, PostgreSQL, Nginx, Linux, Node.js, JWT (HS256), Contabo VPS (Ubuntu)

Darkpearl | 🐙 Github

Jan 2024 - Jul 2025

Founding Engineer

- Engineered a server-side LLM orchestration layer using Vercel AI SDK with multi-provider support (OpenAI, Anthropic, Gemini), implementing fire-and-forget streaming with background agent execution and real-time Pocketbase pub/sub delivery.
- Implemented IP-based in-memory rate limiting (100 req/60s window) on LLM endpoints returning RFC-compliant 429 responses with Retry-After headers, alongside per-project agent state tracking with 5-minute stuck-agent auto-recovery.
- Built a self-hosted AI development platform on a SvelteKit + Pocketbase microservice architecture, enabling live Svelte 5 in-browser compilation, hot-reload preview, and one-click production HTML deployment from a single server.

Technologies / Tools Used : TypeScript, Pocketbase, Vercel AI SDK, OpenAI / Anthropic / Gemini APIs, CodeMirror 6, SSE Streaming, Node.js, Vite

SKILLS

Databases :	PostgreSQL, MySQL, Redis, SQLite, S3
Frameworks & Libraries :	Flask, Vite, SvelteKit, Spring, Django, Vue, React Router DOM, JWT (python-jose, PyJWT), Authlib, Next.js, FastAPI, Spring Boot, Svelte, Angular, Node.js
Languages :	English, Hindi, Kannada
Programming Languages :	C++, Java, Python, TypeScript, Rust, JavaScript, SQL, R
Soft Skills :	Communication, Teamwork, Problem Solving, Creativity, Critical Thinking, Time Management, Adaptability, Leadership, Decision Making
Tools & Platforms :	Jenkins, Kubernetes, Docker, GitLab, GitHub, Git, AWS, GCP, CI/CD, Postman, GitHub Actions

EDUCATION

Dayananda Sagar Academy of Technology and Management | Bengaluru India

Dec 2021 - Jul 2025

BE in Information Science and Engineering

CGPA : 8.50/10

- Represented **College at Startup Mahakumbh 2024**; attended PM's speech at Bharat Madapam, New Delhi.
- Organized **Hackathon** events; co-founded **Innovisions Club**.
- Won multiple **hackathons**, including Cascading Craft.
- Awarded Gold Medal from Karnataka state in IT software solution for Business domain; represented at IndiaSkill2024 - New Delhi.
- Honored with Gold Medal at Vidhana Soudha, Bangalore by the Ministry of skills and entrepreneurship